SNEHANKUR MONDAL

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Summary

Computer science postgraduate with a Physics background and hands-on experience in Python development, scientific computing, data analysis, and web application development. Experienced in building technical projects that combine analytical problem-solving with practical software solutions, with a growing focus on AI/ML and data-driven applications. Brings an interdisciplinary perspective, research experience, and leadership development exposure through Christ University's Future Leaders Programme for Christites (FLPC).

Education

Master of Computer Applications (MCA) Christ University, Delhi NCR | 2026–Present

Master of Science (M.Sc.) – Physics University of Calcutta (2024 – 2026)

Bachelor of Science (B.Sc. Hons) – Physics

University of Calcutta (2021 – 2024)

- CGPA: 7.695 | Core CGPA: 7.9 • College Rank: **2nd**

Technical Skills

Programming: Python, C/C++, Java, R.

Web Technologies: HTML, CSS, JavaScript, React, Node.

Libraries: NumPy, SciPy, Matplotlib, tKinter, PyTorch, Pandas, Scikit-Learn, Seaborn, TensorFlow.

Tools & Platforms: MATLAB, VS Code, Linux, LaTeX, Excel, MySQL, Tableau, Origin.

Domains: Computational Physics, Numerical Methods, Data Analysis, AI/ML, Web Development

Projects

Physics Simulator (C): Developed a C-based physics simulator using functions and numerical formulas to calculate harmonic oscillations, gravity, escape velocity, and orbital parameters.

AI/ML Materials Prediction: Research Project: Developing an AI/ML system that predicts suitable material combinations and proportions from desired properties to simplify material selection and testing.

BakeVerse: AI-Integrated Web Project: Developed a bakery-focused web application using HTML, CSS, JavaScript, and AI/ML features to provide an interactive online experience.

Numerical, Statistical & Quantum Physics Projects: Applied numerical, statistical, and quantum-mechanical methods to solve and analyse computational physics problems.

Marine Data Analysis Tool: Developed Python-based workflows for processing, analysing, and visualising marine scientific datasets.

Portfolio Website: Developed a responsive portfolio website using HTML, CSS, and JavaScript to showcase projects and technical skills.

Experience

Research Project / Researcher: Materials Science & Energy Systems | Jadavpur University | 2026–Present

Analyzing experimental/material datasets using Python for processing, visualization, and **computational interpretation.**

Python Developer: Academic Automation Project | Self-initiated | Dec 2025–Present

Developed Python workflows for academic and Marine Science Research Data processing, numerical estimation, and scientific computation.

Web & Frontend Development Intern | Future Interns | Feb 2026–Mar 2026

Built responsive web interfaces using HTML, CSS, and JavaScript with clean, reusable UI components.

Leadership & Development	**Leadership & Development** Future Leaders Programme for Christites (FLPC) | Christ University • Department-nominated participant, selected to represent the department in a university-wide leadership development programme. • Participated in a structured, one-day workshop featuring leadership-focused activities designed to develop collaboration, communication, and problem-solving skills. Organized college fests and departmental events, managing coordination and execution.
Certifications	• Web Development Fundamentals - **IBM** • Artificial Intelligence Fundamentals - **IBM** • Data Science (Microsoft Certified – DP-900) - **Microsoft** • Full Stack Web Development - **Future Interns** • Data Analytics Certificate – **Google** • Machine Learning Specialization – **DeepLearning.AI & Stanford Online** (In Progress)
Achievements & Awards	• **Best Poster Award** (Postgraduate Category) – MOF-Enhanced PVDF for Triboelectric Energy Harvesting • 2nd Prize – BARC National-Level Quiz • 3rd Prize – Ramakrishna Mission Inter-College Quiz • Represented institution at academic science events
Additional	**Languages:** English, Bengali, Hindi **Interests:** Music, Guitar, Science Outreach, Hackathons(SIH)